

Problem

A capacitor in an RC circuit with $R = 2\ \Omega$ and $C = 4\ \text{F}$ is being charged. The time required for the capacitor voltage to reach 63.2 percent of its steady-state value is:

- (a) 2 s
- (b) 4 s
- (c) 8 s
- (d) 16 s
- (e) none of the above

Step-by-step solution

Step 1 of 2

Consider an RC circuit.

Write the expression for the response of voltage across the capacitor in the circuit.

$$v_c(t) = v_c(\infty) + [v_c(0) - v_c(\infty)]e^{-\frac{t}{\tau}}$$

Here, $v_c(0)$ is the initial value of capacitor voltage, $v_c(\infty)$ is the steady state value of capacitor voltage and τ is the time constant.

The circuit is in steady state at $t < 0$, and the capacitor acts as an open circuit in steady state. Therefore, the voltage across the capacitor for $t < 0$ is,

$$v_c(0) = 0\ \text{V}$$

The time constant of RC circuit is,

$$\tau = RC$$

Substitute $2\ \Omega$ for R and $4\ \text{F}$ for C .

$$\begin{aligned}\tau &= (2\ \Omega)(4\ \text{F}) \\ &= 8\ \text{s}\end{aligned}$$

Step 2 of 2

Consider that the capacitor charges to 63.2 percent of its final or steady-state value. Therefore,

$$v_C(t) = 0.632v_C(\infty)$$

Substitute all the values in the equation of capacitor voltage.

$$v_C(t) = v_C(\infty) + [v_C(0) - v_C(\infty)]e^{-\frac{t}{\tau}}$$

$$0.632v_C(\infty) = v_C(\infty) + [0 - v_C(\infty)]e^{-\frac{t}{8}}$$

$$v_C(\infty)e^{-\frac{t}{8}} = v_C(\infty) - 0.632v_C(\infty)$$

$$e^{-\frac{t}{8}} = 0.368$$

Simplify the expression further to find the value of t .

$$-\frac{t}{8} = \ln(0.368)$$

$$-t = 8(-0.999)$$

$$t \approx 8 \text{ s}$$

Therefore, the time required for the capacitor voltage to reach 63.2% of its steady state value is 8 s.

Hence, the correct option is, c.